Specify a comparator before comparing performance

The shared comparator contract

Information • target and law • units
Source role • ties and constraints

Score tiers

$g = \text{tiers}(u)$

Declare score scale and reference.
Preserve its tie rule and actual grades.
Constructed recipe; no general reporting-loss optimality guarantee.

Reporting optimizer

$g^* \in \arg\min_{g \in \mathcal{F}} R(g)$

Minimize the declared reporting risk.
Exact claims need exact coefficients and the entire feasible family.
Keep its law, objective and restrictions.

Exact-budget selection

$S \subseteq \{1, \ldots, n\}, \quad |S| = b$

Maximize conditional mean utilities summed over the selected subset.
Converting a grade into an allocation requires another explicit rule.

Threshold classification

$a_i \in \{0, 1\}$

Declare the event and both error costs.
The simple cutoff assumes unrestricted binary declarations. Combining flags into grades requires a reporting check.

A new combined report

Criterion cards → midrank sum/max → combined report (constructed).
No automatic joint optimum, protection or inherited grade cap.

Choose fixed τ, matched expected N, or exact G/block-size matching.
Evaluate both co-primary targets.